

CSDS 302 Discrete Math | Fall 2026 | Homework 7

Instructions

Section 9.4: 4, 7, 11, 31,33

Section 9.5: 7, 8, 10, 13be, 16bc, 20

Section 9.8: 3, 15, 18

Section 9.9: 2, 18

(8 points) One blue die and one gray die are rolled, what is the probability that the sum of the numbers is 6, given that both of the numbers are odd?

(8 points) Imagine there is a drug test that is 98% accurate, meaning that 98% of the time, it shows a true positive result for someone using the drug, and 98% of the time, it shows a true negative result for nonusers of the drug. Next, assume 0.5% of people use the drug. If a person selected at random tests positive for the drug, what is the probability the person is actually a user of the drug?

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Feedback

Score: 100%

Good work.

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9.4: 4, 7, 11, 31, 33

(4) 26 letters

$$26 \times 26 = 676$$

One out of 676 will have the same initials; there will be 2 people w/ same first & last initials

(7) $S = \{3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

6 integers at random

$\{3, 12\}$ $\{4, 11\}$ $\{5, 10\}$ $\{6, 9\}$ $\{7, 8\}$
 a_1 a_2 a_3 a_4 a_5

Pigeonhole: More pigeons than pigeonholes therefore there will be 2 pigeons in the same hole which gives the sum of 15.

(11) $n+1$ integers $A = \{1, 2, 3, \dots, 2n\}$ where n is positive int, at least 1 of them must be odd

$$|A| = 2n \quad n+1 > \frac{2n}{2}$$

∴ Even when all n integers picked are even, the $(n+1)^{\text{th}}$ integer will have to be odd

(31) Assume at least 3 has ≤ 2

$$3 \cdot 2 + 2 \cdot 4 = 6 + 8 = 14 \quad \text{(4)}$$

A E FA mx

(33) $A = \{x \mid x \in \mathbb{Z}^+, x < 15\}$
 $2^6 = 64 \Rightarrow 63 \text{ non-}\emptyset \text{ sets}$

k smallest:

$$k + 10 + 11 + 12 + 13 + 14 = k + 60 \Rightarrow 4 \text{ #'s}$$

9.5: 7, 8, 10, 13, 16, 20

(7) a. $\frac{15!}{(13-7)!7!} = 1716$

b. 7 women 6 men

(i) 4W 3M

$$\frac{7!}{(7-4)!4!} \cdot \frac{6!}{(6-3)!3!} = 700$$

(ii) ≥ 1 M:

$$1716 - 1 = 1715$$

(iii) ≤ 3 W

$$\frac{7!}{(7-3)!3!} \cdot \frac{6!}{(6-4)!4!} + \frac{7!}{(7-2)!2!} \cdot \frac{6!}{(6-5)!5!} + \frac{7!}{(7-1)!1!} \cdot \frac{6!}{6!} = 658$$

(c) $\frac{2!}{11!} \cdot \frac{11!}{5!6!} = 1254$

(d) $\frac{2!}{2!} \cdot \frac{11!}{(11-5)!5!} + \frac{11!}{(11-7)!7!} = 792$

(8) a. $\frac{14!}{10!4!} = \frac{14 \times 13 \times 12 \times 11}{4 \times 3 \times 2} = 1001$

b. 6P 8NP

(i) $\frac{6!}{4!2!} \cdot \frac{8!}{6!2!} = 420$ (ii) all

(iii) $\frac{6!}{3!3!} \cdot \frac{8!}{7!} + \frac{6!}{4!2!} \cdot \frac{8!}{8!} = 175$

(c) $1001 - \frac{12!}{8!4!} = 1001 - 495 = 506$

(d) $\frac{2!}{2!} \cdot \frac{12!}{10!2!} + \frac{2!}{2!} \cdot \frac{12!}{8!4!} = 521$

(10) 60 mice; $A = B = 22$; control = 16

$$\frac{60!}{(60-22)!22!} + \frac{38!}{(38-22)!22!} + \frac{16!}{16!} = 1.41543 \times 10^{16}$$

(13) b: $\frac{10!}{5!5!} = 252$
e: $\frac{10!}{9!} + \frac{9!}{9!} = 11$

(14) b: $\frac{40!}{35!5!} - \frac{37!}{32!5!} = 222111$

c: $\frac{222111}{\left(\frac{40!}{35!5!}\right)} = \frac{222111}{658008} \cdot 100\% = 33.8\%$

(20) MILLIMICRON $n=11$

a: $\frac{11!}{2!2!3!} = 146,320$

$\begin{matrix} -2m \\ 3I \\ 2L \\ 1C \\ 1R \\ 1O \\ 1N \end{matrix}$

b: $\frac{9!}{2!3!} = 30240$ c: $\frac{9!}{2!3!2!} = 15120$

q. 8: 3, 15, 18

(3) $A \cup B \cup C = S$ $p(A)=0.4$ $p(B)=0.2$

a: $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.6$

b: $P(C) = 0.2$ NO.

$A \cup B \cup C = S$

$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(B \cap C) - P(A \cap C)$

$P(C) = 1 - 0.6 + P(B \cap C) + P(A \cap C)$

$P(C) = 0.4 + P(B \cap C) + P(A \cap C) > \underline{0.2}$

(15) Price: \$40,000 $P: \$20$

Add: \$1000, \$500

$\frac{1}{8000}$ (gross price + TV + coupon - 2 (net price))

$\frac{1}{8000} (39980 + 980 + 480 - 2(2980))$

= \$11.82

(18) 1, 2, 2', 8, 8'

1 2 2'	1 2' 8	2 2' 8	2 8 8'
1 2 8	1 2' 8'	2 2' 8'	2' 8 8
1 2 8'	1 8 8'		

SUM 5 - 1 - 0.1

SUM 11 - 4 - 0.4

SUM 17 - 1 - 0.1

SUM 12 - 2 - 0.2

SUM 18 - 2 - 0.2

$5 \cdot 0.1 + 11 \cdot 0.4 + 17 + 12 \cdot 0.2 + 18 \cdot 0.2$

= 12.6

q. 9: 2, 18

(2) $P(X|Y) = \frac{1}{3}$ $P(Y) = \frac{1}{4}$

$P(X|Y) = \frac{P(X \cap Y)}{P(Y)}$

$P(X \cap Y) = P(X|Y) \cdot P(Y) = \frac{1}{3} \cdot \frac{1}{4} = \frac{1}{12}$

(18) $P(A \cap B) = P(A) \cdot P(B)$, $P(A) \neq 0$, $P(B) \neq 0$

$P(A|B) = P(A)$ and $P(B|A) = P(B)$

Using $P(A|B) = \frac{P(A \cap B)}{P(B)}$ then:

$P(A \cap B) = P(B) \cdot P(A|B) = P(A) \cdot P(B)$

$P(A|B) = P(A)$ and if $P(B|A) \cdot P(A)$

= $P(B \cap A) = P(A) \cdot P(B)$

$\Rightarrow P(B|A) = P(B)$

QI. Both # are odd, sum 6

1B 1G
3B 3G
5B 5G

$$\frac{3}{9} = \frac{1}{3}$$

pairs: 4 (1,1), (1,3), (1,5),
(3,1), (3,3), (3,5),
(5,1), (5,3), (5,5)

QII. 98% accurate (TP or TN)
0.5% use drug

$P(\checkmark)$ given $P(TP)$

$$P(\text{druggy} | \text{Pos}) = \frac{P(\text{Pos} | \text{druggy}) \cdot P(\text{druggy})}{P(\text{Pos})}$$

$$P(\text{Pos}) = 0.98(0.005) + 0.02(0.995)$$

$$P(\text{Pos}) = 0.0248$$

$$P(\text{druggy} | \text{Pos}) = \frac{0.8 \cdot 0.005}{0.0248} = 16\%$$